

2. Safety Boundary, Risk Assessment, Stop Conditions & Signed Checkpoints

2.1 Non-Negotiable Project Safety Boundary

To maintain absolute electrical and physical safety, all team members must strictly comply with the following mandatory safety boundaries:

- Closed-Case Requirement: The metal enclosure of the source ATX PSU must remain completely closed at all times. No covers shall be removed, and no tools, probes, screws, or loose wires may be inserted through ventilation openings.
- De-energized Construction: All wiring changes, component additions, and terminal adjustments must be performed with the main AC power cord physically disconnected from the wall inlet.
- +5VSB Live Awareness: +5VSB (Pin 9) is continuously energized whenever AC power is connected, even when the main rails (+3.3V, +5V, +12V, -12V) are commanded off via **PS_ON**.
- Control Integrity: **PS_ON** (Pin 16) must be controlled solely using an insulated, maintained switch connected to **COM**. Never use an exposed paperclip or loose jumper wire.
- Branch Fuse Protection: Every accessible positive output branch must be individually insulated and protected by an instructor-approved fuse positioned before any exposed output terminal or wiring. Internal PSU protection does not substitute for branch protection.

- Thermal Safety: Never hold power resistors or components by hand during operation, and never use touch as a thermal measurement method

2.2 Project Risk Assessment & Mitigation Matrix

Hazard	Severity	Initial Cause	Preventive Control
High Current Output Short Circuit	High	Accidental contact between live positive rail terminals and COM or enclosure.	Install separate fast-acting fuses in each output branch
Electric Shock	High	Opening the PSU metal case or probing inside ventilation slots.	Keep source PSU case sealed
Overheating	Medium	Overloading output rails, poor ventilation, or excessive loss in buck-boost converter.	Enforce maximum current limits

Loose Strand	Medium	Poor wire crimping or untrimmed multi-strand copper wires at terminals.	Use insulated wire ferrules and heat-shrink tubing on all crimped connections
Reverse Polarity	Medium	Misinterpreting ATX 24-pin connector orientation	Verify all pin numbers using a DMM continuity test and manufacturer datasheet prior to first power-on.

2.3 Stop-Work Conditions

All operations, measurements, and testing must **STOP IMMEDIATELY** if any of the following conditions occur:

1. The connector orientation, pin assignment, or signal function is unclear.
2. Any component, enclosure, cable insulation, fuse holder, or binding post is damaged or missing.
3. Any rail output is missing, unstable, or repeatedly cycling on and off.
4. Measured current, voltage ripple, or component temperature exceeds approved safety limits.
5. Unusual smell, smoke, abnormal sound, or excessive heat generation is detected.

6. Oscilloscope reference/ground connections are ambiguous (risking short-to-ground through probe earth).
7. An official instructor checkpoint sign-off has not been completed prior to the next step.

Emergency Shutdown Procedure: Safely command **PS_ON** off using the insulated switch. Unplug the AC power cable without touching suspected live wiring. Warn team members, allow stored energy/heat to dissipate, and report immediately to the laboratory instructor.